Teaching notes

Make 4 copies of each information sheet.

Give one information sheet to each pair of students. They should read and summarise the information on their sheet using pictures and a word limit of 12 words. Students then circulate around the room explaining their poster and finding out about three other threats to insect pollinators.

During this time keywords/concepts could be added to the questions on the whiteboard.

Finally, students should get into groups of 4 and answer the questions on the group question sheet.

The extension task questions could be used as part of a group discussion.

Additional teaching notes

This resource can be used as a standalone activity or with the Teachit Science resources 'Wind or insect pollination' (22853) and 'Insect pollinators' (22854) to make a complete lesson.

If using the resources together then see below for suggested learning outcomes:

- name four insect pollinators
- outline one consequence of a decrease in the numbers of insect pollinators on our food supply
- describe four threats to abundance of honey bees and other insect pollinators
- for two of these threats explain how they could be reduced.

The learning outcomes could be shared with students at the start of the activity and left on a whiteboard during the activity. Keywords and phrases, 'memory-joggers' could be added by students during the activity.

Background information

http://butterfly-conservation.org/88/biodiversity-action-plans.html

www.bbka.org.uk/learn/general_information/life_in_the_**hive**

http://www.bbc.co.uk/news/science-environment-23351215

www.operaresearch.eu

http://www.parliament.uk/briefing-papers/POST-PN-348/insect-pollination-january-2010

Suggested answers

Group question sheet

1. Our staple foods, wheat, rice etc are wind pollinated. If we lost a large number of insect pollinators we would still have plenty to eat but we would lose foods which provide us with essential micro-nutrients, vitamins and minerals, antioxidants and variety in our diet.
2. Through wind pollination or by insects and other animals.
3. Honeybees, bumblebees, hoverflies, butterflies and moths
4. The term 'agricultural intensification' describes what has happened to farming practices over the last 60 years. Only one type of plant is grown in each field. Herbicides are used to kill any other plant which may grow on the land. Even when the crop is grass e.g. for silage or grazing, only a few species will be found growing in the field and almost no flowering plants. For many farmers this is the most efficient method of production.
5. Reduced the availability of flowering plants and nesting sites for insects, birds and mammals. The UK has lost 97% of its wildflower meadows since the 1930s. The countryside has been stripped of the wild flowers which sustained populations of bees, butterflies etc.
6. Each farm needs to have natural, wildflower rich areas to support bumblebees, butterflies and other insect pollinators when their crops are not in flower. Or, farmers could 'import' honeybees and their hives when they need them.
7. Varroa mite
8. Colonies of bumblebees should be screened and certified disease free before they are imported into this country.
9. Bumblebees exposed to these controversial pesticides collect just half the pollen they would otherwise harvest, depriving their growing young of their only source of protein.

   It would allow more independent research to be done to ascertain the exact effect these pesticides have on wild insect pollinators.
10. Bees and butterflies are particularly sensitive to wet, cool, windy conditions. Cold wet summers can have a detrimental effect on their numbers and breeding success. They may move northwards as the climate warms and new species could become established in the southern counties.

Suggested answers

Task – extension

1. To ensure the results are reliable, unbiased and can be trusted by decision makers and the public.
2. Farmers/landowners earn their livelihood from the land. Many need to farm intensively to make a living. If we value a rich, diverse natural landscape farmers need to be paid to maintain it.
3. Although at the top of many food webs we are sustained by the organisms below us in that web. Damaged habitats/environments cannot always be repaired. We don't know how much damage/loss can be endured before it has a detrimental effect on us.

The natural world is our heritage and we should pass it on to future generations in good order.

People need green spaces.

Extinction is quick; evolution is slow

Group question sheet

Answer the following questions.

1. Imagine we lost all our insect pollinators. What effect would this have on our diets?

.....

.....

.....

.....

2. How is pollen transferred from one plant to another?

.....

.....

3. Which insects play an important role in pollinating our crops?

.....

.....

4. What does the term ‘agricultural intensification’ mean?

.....

.....

5. Describe how intensive agriculture affects wild insect pollinators?

.....

.....

.....

6. What can be done to ensure there are enough insects to pollinate flowering crops?

.....

.....

.....

7. Give an example of a pest or disease which affects honeybees.

.....

.....

8. How can we prevent imported colonies of bumblebees from spreading diseases to our wild bees?

.....

.....

9. Explain why neonicotinoid insecticides have been given a two-year ban by the EU.

.....

.....

.....

10. Outline the role climate change might play in the abundance and distribution of butterflies.

.....

.....

.....



Task – extension

1. Why do you think it might be important for research into the possible detrimental effects of pesticides to be carried out by independent organisations? (Independent organisations do not receive money for their research from anybody who has a vested interest in the outcome of the research).

.....

.....

.....

.....

.....

.....

2. Whose responsibility is it to maintain the natural diversity of our countryside?

   Who should pay - farmers, landowners, the government or the public?

.....

.....

.....

.....

.....

.....

3. How important is it that the biodiversity of the UK is maintained?

   Why might it be important to our own health and well-being?

.....

.....

.....

.....

.....

.....

Information sheet 1

Agricultural intensification

The term ‘agricultural intensification’ describes what has happened to farming practices over the last 60 years. Only one type of plant is grown in each field. Herbicides are used to kill any other plant which may grow on the land. Even when the crop is grass e.g. for silage or grazing, only a few species will be found growing in the field and almost no flowering plants. For many farmers this is the most efficient method of production.

However, agricultural intensification has reduced the availability of flowering plants and nesting sites for insects, birds and mammals. The UK has lost 97% of its wildflower meadows since the 1930s. Such changes affect wild insect pollinators because they are totally dependent on wild flowers growing in the landscape.

Flowering crops such as oilseed rape produce a super abundance of flowers over a very short period of time. For there to be enough insects in the environment to pollinate the flowering oilseed rape each farm needs to have natural, wildflower rich areas to support bumblebees, butterflies and other insect pollinators when their crops are not in flower.

To overcome this problem in the USA, large numbers of honeybees and their hives are transported around the country. The bees are taken to orchards as the trees come into blossom. When the fruit has set the hives are moved to newly flowering orchards. In recent years many hives in the USA have suffered the devastating colony collapse disorder (CCD) in which all the adult bees simply disappear from the hive. CCD may be linked to the practice of transporting bees over large distances or to pesticides.

Information sheet 2

Pests and diseases

Pests and diseases are the main threat to honeybees, particularly an introduced parasitic mite called Varroa. Varroa infestations can be treated but the mite is beginning to show resistance to some treatments.

Most beehives in Britain are managed by amateur beekeepers in their spare time. Beekeeping associations provide training but the quality depends on the expertise of local volunteers.

The impact of pests and diseases on other pollinating insects has not been well studied, but is probably less severe than on honeybees as they are particularly exposed to new diseases spread by human activity. Bumblebees may be affected by disease transfer from imported bumblebees that have been released in glasshouses and polytunnels. This was linked to severe declines in some American bumblebee species, but has not been investigated in the UK. To prevent this happening colonies of bumblebees should be screened and certified disease free before they are imported into this country.

Treating Varroa and other health problems effectively and breeding more resistant bees are important long term goals. Meanwhile, the standard of beekeeping could be improved through passing the results of scientific research to beekeeping associations and encouraging beekeepers to keep up to date with developments.

Information sheet 3

Agrochemicals

Insecticides have been developed to kill pest species of insects but they can also be toxic to our wild insect pollinators. EU legislation does not require risk assessments for wild pollinators. Measures to protect honeybees, such as restricted spraying times, may not protect other insect pollinators that are active at different times of the day. Therefore, risks to wild pollinators remain difficult to assess. Herbicides and fertilisers may indirectly affect pollinators by reducing wildflower abundance and diversity.

For several years some conservation groups have been calling for a ban on neonicotinoid pesticides. These are "systemic" pesticides, this means they are absorbed and transported throughout the plant.

Bumblebees exposed to these controversial pesticides collect just half the pollen they would otherwise harvest, depriving their growing young of their only source of protein.

This research has been hailed as important by independent scientists because it sheds light on how the neonicotinoid pesticides can harm bees.

"Pollen is the only source of protein that bees have, and it is vital for rearing their young," said Professor Dave Goulson, at the University of Sussex and who led the study. "Collecting it is fiddly, slow work for the bees and intoxicated bees become much worse at it. Without much pollen, nests will inevitably struggle."

A two-year EU ban of three neonicotinoids, the most widely used insecticides in the world, began in December 2013. This ban should allow more independent research to be done to ascertain the exact effect these pesticides have on wild insect pollinators.

Information sheet 4

Climate Change

There is uncertainty about the potential impact of climate change on UK insect pollinators. Pollinators that can thrive in a wide range of habitats, like honeybees, may adapt. However, there could be a mismatch between flowering dates of food plants and emergence dates of pollinators. The flowering of blackcurrants and the emergence of its insect pollinators have diverged by 28 days since the 1970s. This could expose pollinators to periods of starvation and to poor fruit setting in the crop plant.

Bees and butterflies are particularly sensitive to wet, cool, windy conditions. Cold wet summers can have a detrimental effect on their numbers and breeding success. They may move northwards as the climate warms and new species could become established in the southern counties. To reduce the effects of climate change land owners and managers could establish corridors of natural land which would connect together different habitats. These would enable species to move easily from one area to another.